

Ergodicity and transducers

Marie COUTEREEL SLEGHEM, University Paris-Cité, IRIF,
joint work with Olivier CARTON and Matthieu PICANTIN

Highlights, September 2026, Vienna

Motivation

Group theory difficult problems :

Motivation

Group theory difficult problems :

- Burnside problem
- Milnor problem

Motivation

Group theory difficult problems :

- Burnside problem
 - Milnor problem
- } The Grigorchuk automaton solves both of them !

Motivation

Group theory difficult problems :

- Burnside problem
 - Milnor problem
- } The Grigorchuk automaton solves both of them !

Group theory open problems :

Motivation

Group theory difficult problems :

- Burnside problem
 - Milnor problem
- } The Grigorchuk automaton solves both of them !

Group theory open problems :

deciding transitivity

Motivation

Group theory difficult problems :

- Burnside problem
 - Milnor problem
- } The Grigorchuk automaton solves both of them !

Group theory open problems :

deciding transitivity $\implies$ deciding ergodicity

Ergodicity

Definition (Ergodicity)

A function $f : E \rightarrow E$ is ergodic if:

- it preserves the measure

$$\mu(f^{-1}(X)) = \mu(X)$$

- it has **no** (non-trivial) stable subset

$$f^{-1}(X) \subset X \implies \mu(X) = 0 \text{ or } \mu(X) = 1$$

Ergodicity

Intuitively, a function is ergodic if it shuffles well.

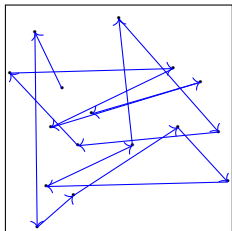


Figure: Ergodic map on $[0, 1]^2$

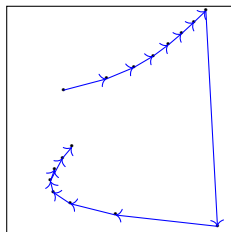
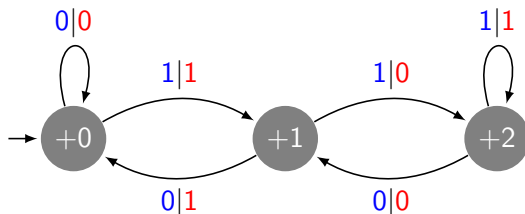


Figure: Non-ergodic map on $[0, 1]^2$

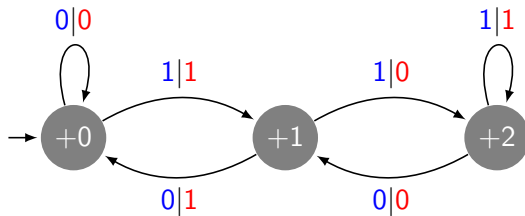
Ergodicity

Is the function generated by a transducer ergodic ?



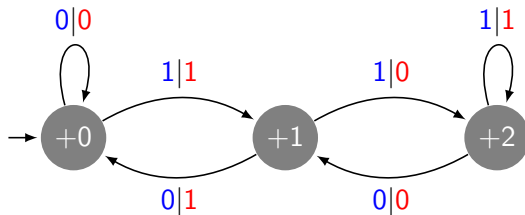
Transducers

Multiplication by 3 for 2-adics numbers



Transducers

Multiplication by 3 for 2-adics numbers

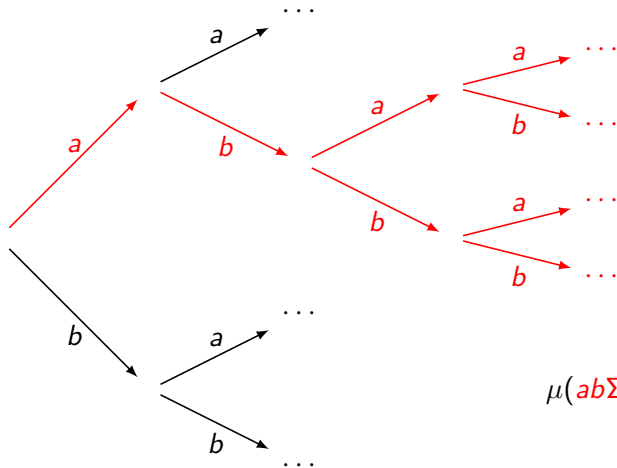


...0001011

...0100001

Uniform Measure

On Σ^ω , we take the uniform measure such that $\mu(x_0 \cdots x_{n-1} \Sigma^\omega) = |\Sigma|^{-n}$.



$$\mu(ab\Sigma^\omega) = 2^{-2}$$

Deciding Ergodicity

Is the function generated by a transducer ergodic ?

Deciding Ergodicity

Is the function generated by a transducer ergodic ?

Theorem [LIPINA 2020]

If a function is computed by a transducer such that :

- the transducer is **lettre-to-letter**
- the transducer is **deterministic**
- the function **preserves the prefixes**,
ie $f(u\Sigma^\omega) \subset v\Sigma^\omega$, with $|u| \leq |v|$
- the alphabet is of size 2,

then the ergodicity is decidable.

Deciding Ergodicity

Is the function generated by a transducer ergodic ?

Theorem [LIPINA 2020]

If a function is computed by a transducer such that :

- the transducer is **lettre-to-letter**
- the transducer is **deterministic**
- the function **preserves the prefixes**,
ie $f(u\Sigma^\omega) \subset v\Sigma^\omega$, with $|u| \leq |v|$
- the alphabet is of size 2,

then the ergodicity is decidable.

Objective: Generalise this result.

Deciding Ergodicity

Is the function generated by a transducer ergodic ?

Theorem [LIPINA 2020]

If a function is computed by a transducer such that :

- the transducer is **letter-to-letter**
- the transducer is **deterministic**
- the function **preserves the prefixes**,
ie $f(u\Sigma^\omega) \subset v\Sigma^\omega$, with $|u| \leq |v|$
- the alphabet is of size 2,

then the ergodicity is decidable.

Objective: Generalise this result.

What does a transducer that computes an ergodic **prefix-preserving** function look like?

Deterministic condition

Is there non-determinisable transducer that computes such functions ?

Deterministic condition

Is there non-determinisable transducer that computes such functions ?

Proposition [CS 2026]

If the function generated by a transducer

- preserves the measure
- preserves the prefixes

then it is generated by a (computable) deterministic transducer.

Letter-to-letter condition

Is there deterministic non letter-to-letter transducer that computes such functions ?

Letter-to-letter condition

Is there deterministic non letter-to-letter transducer that computes such functions ?

Proposition [CS 2026]

If the function generated by a deterministic transducer

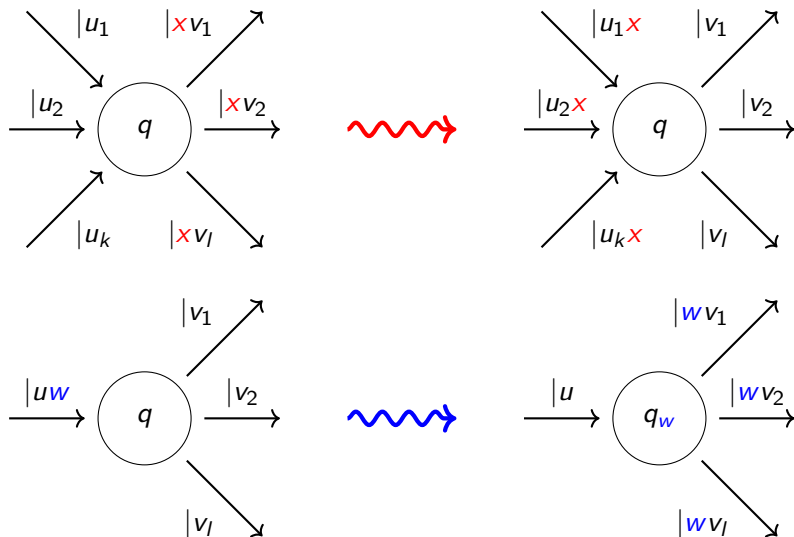
- preserves the measure
- preserves the prefixes

then it is generated by a (computable) deterministic letter-to-letter transducer.

Lemma

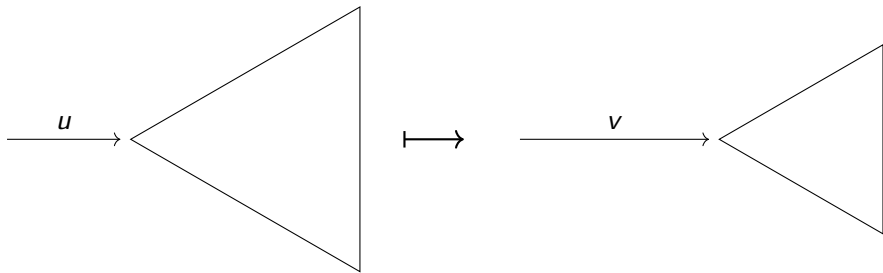
If a function generated by a transducer preserves the prefix, then it is generated by an ϵ output transitions free transducer.

Letter-to-letter condition



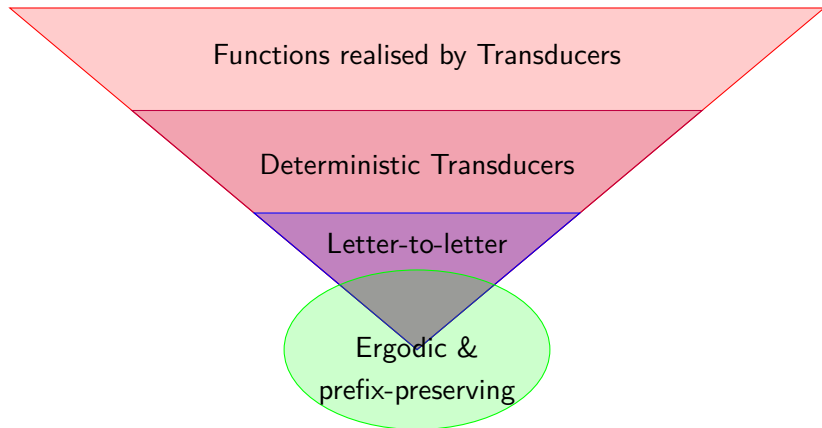
Letter-to-letter condition

If $q_0 \xrightarrow{u|v} q$, with $|u| < |v|$



then it does not preserve the measure !

Recap



Theorem [LIPINA 2020]

If a function is computed by a transducer such that :

- the transducer is **lettre-to-letter**
- the transducer is **deterministic**
- the function **preserves the prefixes**
- the alphabet is of size 2,

then the ergodicity is decidable.

Theorem [LIPINA 2020][CS 2026]

If a function is computed by a transducer such that :

- the transducer is ~~lettre-to-letter~~
- the transducer is ~~deterministic~~
- the function **preserves the prefixes**
- the alphabet is of size 2,

then the ergodicity is decidable.